View Official Syllabus Source

REVISION 2021 | SEMESTER 2

Engineering Chemistry

Course Code: 2072 | Credits: 3 | L:2 T:1 P:0

Course Overview

This course is designed for Diploma students in **Chemical Engineering** and **Food Processing Technology**. It bridges the gap between basic scientific principles and industrial applications. The curriculum covers inorganic trends, analytical precision, organic reaction mechanisms, and the physical chemistry of kinetics and equilibrium.

Teaching & Assessment Scheme

Component	Details
Contact Hours	45 Periods per semester
Category	Engineering Science
Prerequisites	Applied Chemistry (Sem 1), Basic Physics & Maths

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Course Outcomes (COs)

- **CO1:** Outline the basic concepts of inorganic and analytical chemistry.
- **CO2:** Comprehend fundamental concepts of organic chemistry to solve elementary problems.
- **CO3:** Summarise preparation, properties, and uses of industrially important organic compounds.
- **CO4:** Apply chemical equilibria and kinetics for chemical engineering applications.

Module 1: Inorganic and Analytical Chemistry

1.1 Periodicity in Properties

Modern Periodic Law: The physical and chemical properties of elements are periodic functions of their atomic numbers. This law, proposed by Moseley, led to the development of the Modern Periodic Table.

Periodic Trends Down a Group and Along a Period

Property	Along a Period (Left to Right)	Down a Group (Top to Bottom)
Atomic Radius	Decreases	Increases
Ionization Energy	Increases	Decreases
Electron Affinity	Increases	Decreases
Electronegativity	Increases	Decreases

> **Malayalam Note:** [illegible] (Atomic Radius) [illegible], [illegible].

1.2 Analytical Methods

Analytical chemistry involves qualitative and quantitative analysis. Key concepts include:

- **Solubility Product (K_{sp}):** The product of the molar concentrations of ions in a saturated solution.
- **Common Ion Effect:** The suppression of the degree of dissociation of a weak electrolyte by the addition of a strong electrolyte containing a common ion.

Calculation: Mean and Standard Deviation

Data: 10.1, 10.2, 10.3

Mean ($\bar{x}$): $(10.1 + 10.2 + 10.3) / 3 = 10.2$
Standard Deviation (s): $\sqrt{[\Sigma(x_i - \bar{x})^2 / (n-1)]} = 0.1$

1.3 Chromatography

Chromatography is a technique for separating mixtures. It involves a stationary phase and a mobile phase.

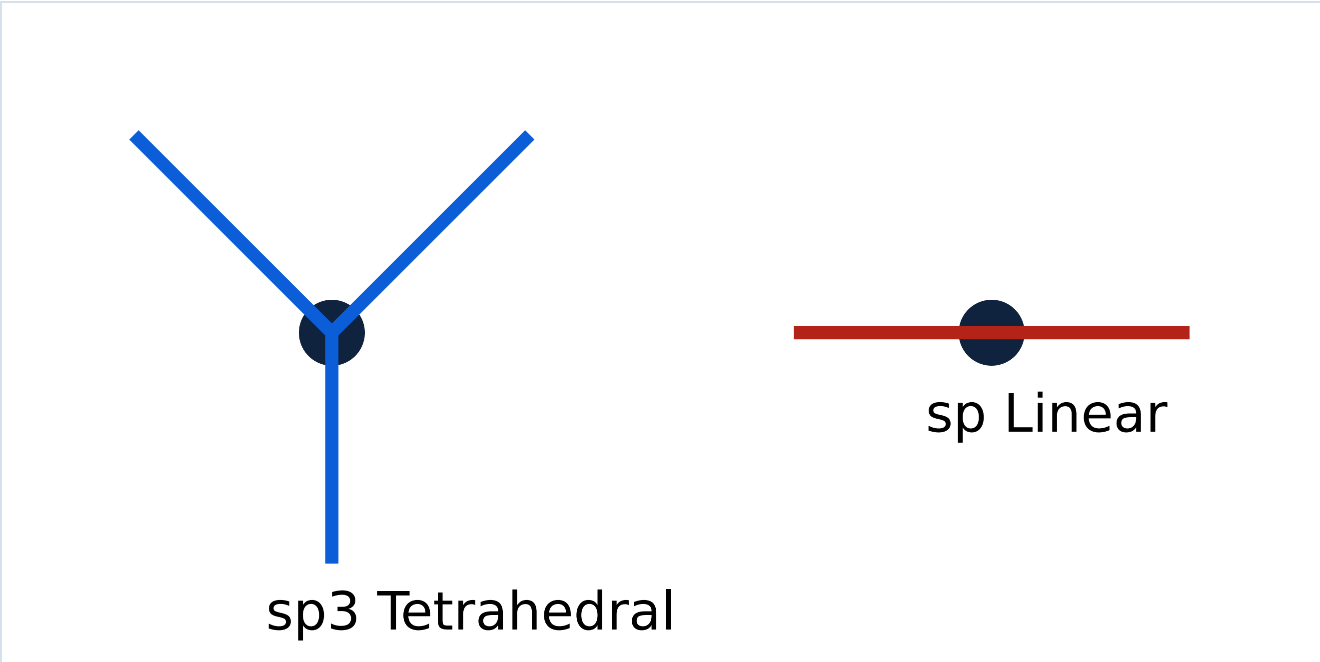
- **Adsorption Chromatography:** Based on differential adsorption (e.g., Column Chromatography, TLC).
- **Partition Chromatography:** Based on differential partitioning between phases (e.g., Paper Chromatography).

Module 2: Fundamental Concepts of Organic Chemistry

2.1 Bonding and Hybridization

Carbon undergoes hybridization to form stable bonds:

- **sp^3:** Tetrahedral (e.g., Methane, Ethane). 109.5° angle.
- **sp^2:** Trigonal planar (e.g., Ethene, Benzene). 120° angle.
- **sp:** Linear (e.g., Ethyne). 180° angle.



Simplified geometry of sp3 and sp hybridization.

2.2 Qualitative Analysis

Lassaigne's Test: Used to detect Nitrogen, Sulphur, and Halogens in organic compounds by fusing the compound with Sodium metal.

2.3 Organic Reactions

- **Homolytic Fission:** Forms free radicals.
- **Heterolytic Fission:** Forms carbocations (C+) or carbanions (C-).
- **Nucleophiles:** Electron-rich species (e.g., OH^-).
- **Electrophiles:** Electron-deficient species (e.g., NO_2^+).

Module 3: Industrially Important Organic Compounds

3.1 Hydrocarbons

Wurtz Reaction: Preparation of higher alkanes from alkyl halides using Sodium in dry ether.

$$2RX + 2Na \rightarrow R\text{-}R + 2NaX$$

Markovnikov's Rule: In the addition of HBr to an unsymmetrical alkene, the negative part (Br-) attaches to the carbon with fewer hydrogen atoms.

3.2 Alcohols and Phenols

- **Ethanol:** Prepared by fermentation of molasses. Used as a solvent and fuel.
- **Phenol:** Prepared via **Dow's Process** (from chlorobenzene) or **Cumene Process**.

3.3 Glucose Structure

Glucose is an aldohexose. It exists in open-chain and cyclic forms. Reactions include acetylation (proves 5 -OH groups) and osazone formation.

3.4 Amino Acids

Zwitterion: A dipolar ion formed by the internal transfer of a proton from the carboxyl group to the amino group.

> **Malayalam Note:** [illegible] Zwitterion [illegible].

Module 4: Chemical Equilibria and Kinetics

4.1 Chemical Equilibrium

Law of Mass Action: The rate of a chemical reaction is proportional to the product of the active masses (molar concentrations) of the reactants.

$$K_p = K_c(RT)^{\Delta n}$$

Le-Chatelier's Principle: If a system at equilibrium is subjected to a change in concentration, temperature, or pressure, the equilibrium shifts in a direction that tends to undo the effect of the change.

4.2 Chemical Kinetics

First Order Reaction: The rate depends on the concentration of only one reactant.

```
k = (2.303 / t) log(a / (a-x))
```

First-Order Rate Constant Calculator

Initial Conc (a): 100

Final Conc (a-x): 50

Time (t): 10

Calculate k

Result: --

Formula Bank

Analytical Errors:

Absolute Error = |True Value - Observed Value|

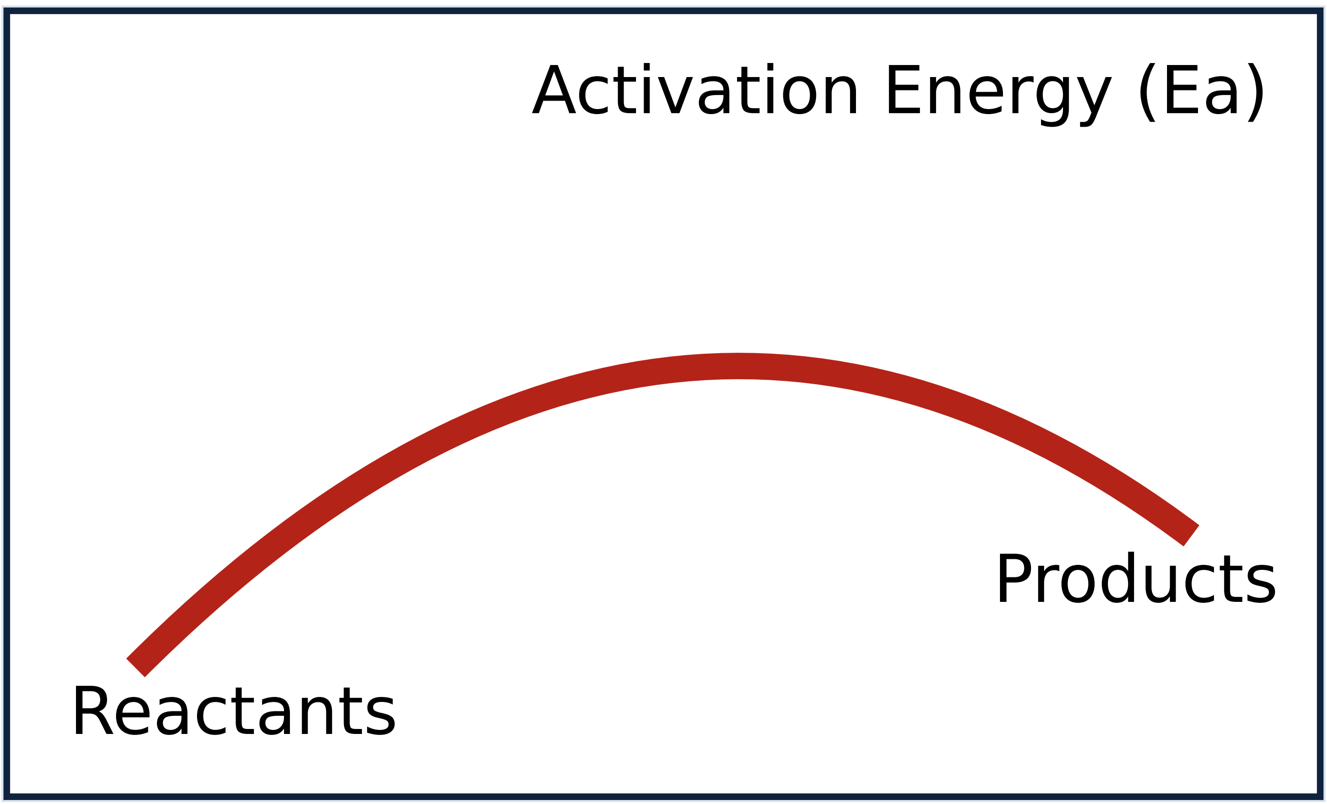
Relative Error = (Absolute Error / True Value) × 100

Kinetics:

Half life ($t_{1/2}$) for 1st order = 0.693 / k

Arrhenius Eq: $k = A\ e^{-Ea/RT}$

Visual Library



Reaction Energy Profile showing Activation Energy.

Module-wise Practice Questions

▶ **Q: Define Electronegativity. How does it vary in a period?**

▶ **Q: What is the difference between Order and Molecularity?**

▶ **Q: Explain the application of Le-Chatelier's principle in Haber's process.**

Practice Model Paper

Part A (Short Answer - 2 Marks each)

1. State Modern Periodic Law.
2. Define Zwitterion with an example.
3. What is the unit of rate constant for a first-order reaction?

Part B (Long Answer - 5 Marks each)

1. Explain sp3 hybridization in Methane.
2. Describe the preparation of Phenol by Dow's process.
3. Derive the relationship between Kp and Kc.

Quick Revision

- **Moseley:** Atomic number is the basis of periodic table.
- **Ksp > Ionic Product:** Solution is unsaturated (no precipitation).
- **Lassaigne's Test:** Na + C + N → NaCN (for Nitrogen detection).
- **Markovnikov:** "Rich get richer" (Hydrogen goes to C with more H).
- **Arrhenius:** Rate increases exponentially with temperature.

Glossary

Accuracy:

Closeness of a measurement to the true value.

Precision:

Closeness of agreement between repeated measurements.

Functional Group:

An atom or group of atoms that determines the chemical properties of an organic compound.

[illegible]

[illegible]:

- **[illegible] 1:** [illegible], [illegible], [illegible].
- **[illegible] 2:** [illegible] (Hybridization), [illegible].
- **[illegible] 3:** [illegible], [illegible], [illegible], [illegible].
- **[illegible] 4:** [illegible] (Equilibrium), [illegible] (Kinetics).
